

Model Analysis Report

Model: occft_yolo

Table: accuracies.csv

Metric	Value
accuracy	0.7571
accuracy_top2	0.9571
accuracy_top3	0.9829

Table: precision_recall_f1.csv

Class	Metric	Value
ANGRY	precision	0.8611
ANGRY	recall	0.62
ANGRY	f1_score	0.7209
ANGRY	support	50
DISGUST	precision	0.7391
DISGUST	recall	0.68
DISGUST	f1_score	0.7083
DISGUST	support	50
FEAR	precision	0.75
FEAR	recall	0.42
FEAR	f1_score	0.5385
FEAR	support	50
HAPPY	precision	0.9804
HAPPY	recall	1
HAPPY	f1_score	0.9901
HAPPY	support	50
NEUTRAL	precision	0.6667
NEUTRAL	recall	1
NEUTRAL	f1_score	0.8
NEUTRAL	support	50

SAD	precision	0.6596
SAD	recall	0.62
SAD	f1_score	0.6392
SAD	support	50
SURPRISE	precision	0.7164
SURPRISE	recall	0.96
SURPRISE	f1_score	0.8205
SURPRISE	support	50
macro_avg	precision	0.7676
macro_avg	recall	0.7571
macro_avg	f1_score	0.7454
weighted_avg	precision	0.7676
weighted_avg	recall	0.7571
weighted_avg	f1_score	0.7623

Table: probs_ytrue_ypred.csv

Prob_ANGRY	Prob_DISGUST	Prob_FEAR	Prob_HAPPY	Prob_NEUTRAL	Prob_SAD	Prob_SURPRISE	True_Label	Predicted_Label	Prediction_Confidence
0.4371	0.0006504	0.003338	8.245e-05	0.15	0.4088	1.204e-05	ANGRY	ANGRY	0.4371
0.0001538	0.9998	9.589e-08	8.685e-08	4.128e-08	3.878e-06	1.141e-10	ANGRY	DISGUST	0.9998
0.9182	0.001471	0.000443	2.914e-06	0.008549	0.07128	4.247e-06	ANGRY	ANGRY	0.9182
0.9516	0.0005232	0.005831	1.964e-06	0.02219	0.01982	4.112e-05	ANGRY	ANGRY	0.9516
0.8763	0.0004223	0.0007069	2.635e-06	0.002238	0.1204	8.667e-06	ANGRY	ANGRY	0.8763
0.8755	0.07304	0.005748	0.0001252	0.03846	0.006902	0.0002596	ANGRY	ANGRY	0.8755
0.1212	0.1005	0.00012	5.616e-05	0.02044	0.7576	3.004e-06	ANGRY	SAD	0.7576
0.3499	0.02133	0.02005	1.318e-05	0.02451	0.5841	4.152e-05	ANGRY	SAD	0.5841
0.1111	0.01458	0.01269	8.531e-06	0.003999	0.8576	2.309e-05	ANGRY	SAD	0.8576
0.5794	0.002103	0.00111	1.35e-05	0.0156	0.4018	1.031e-06	ANGRY	ANGRY	0.5794

Images from confusion_matrix:

Image: occft_yolo_cm.png

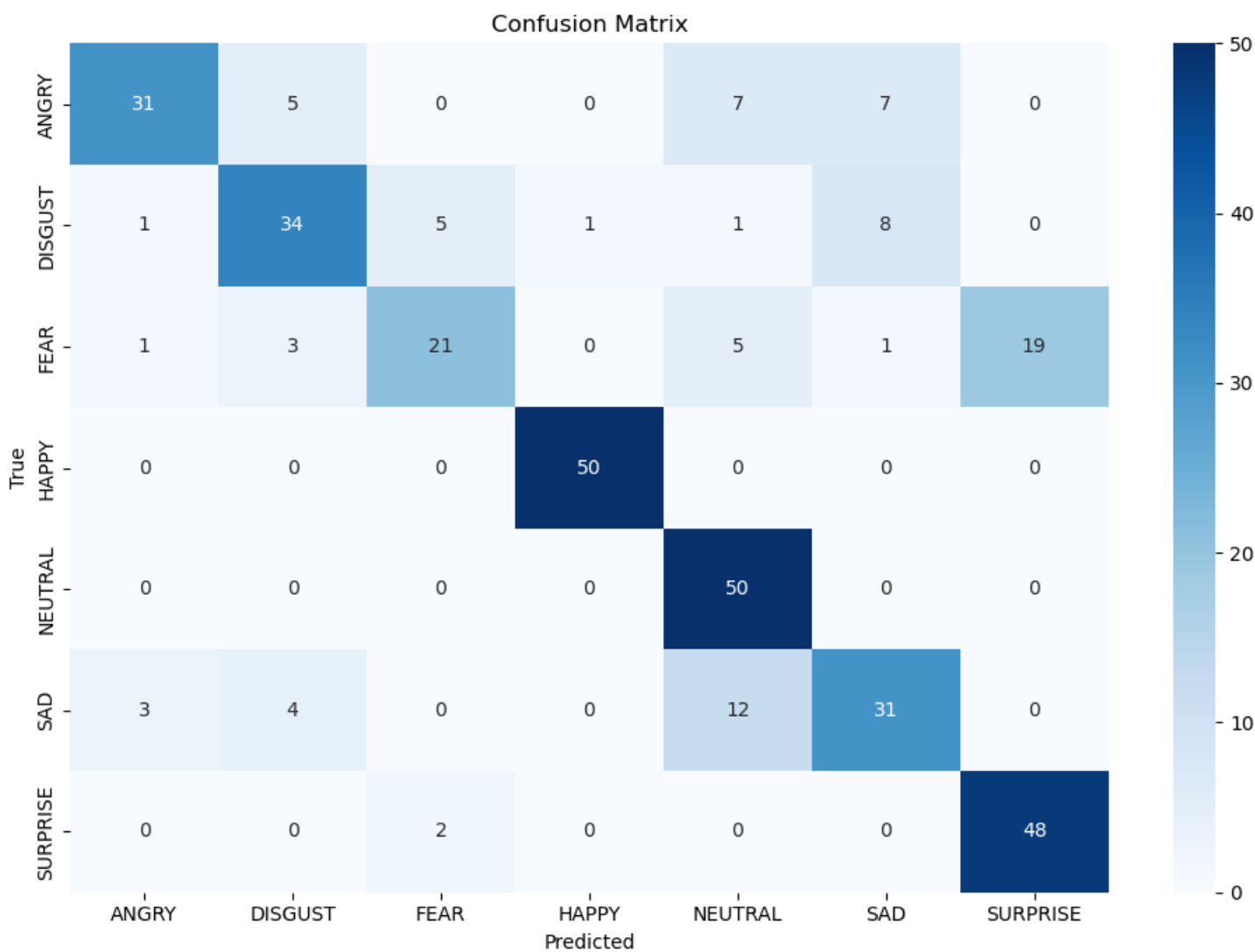
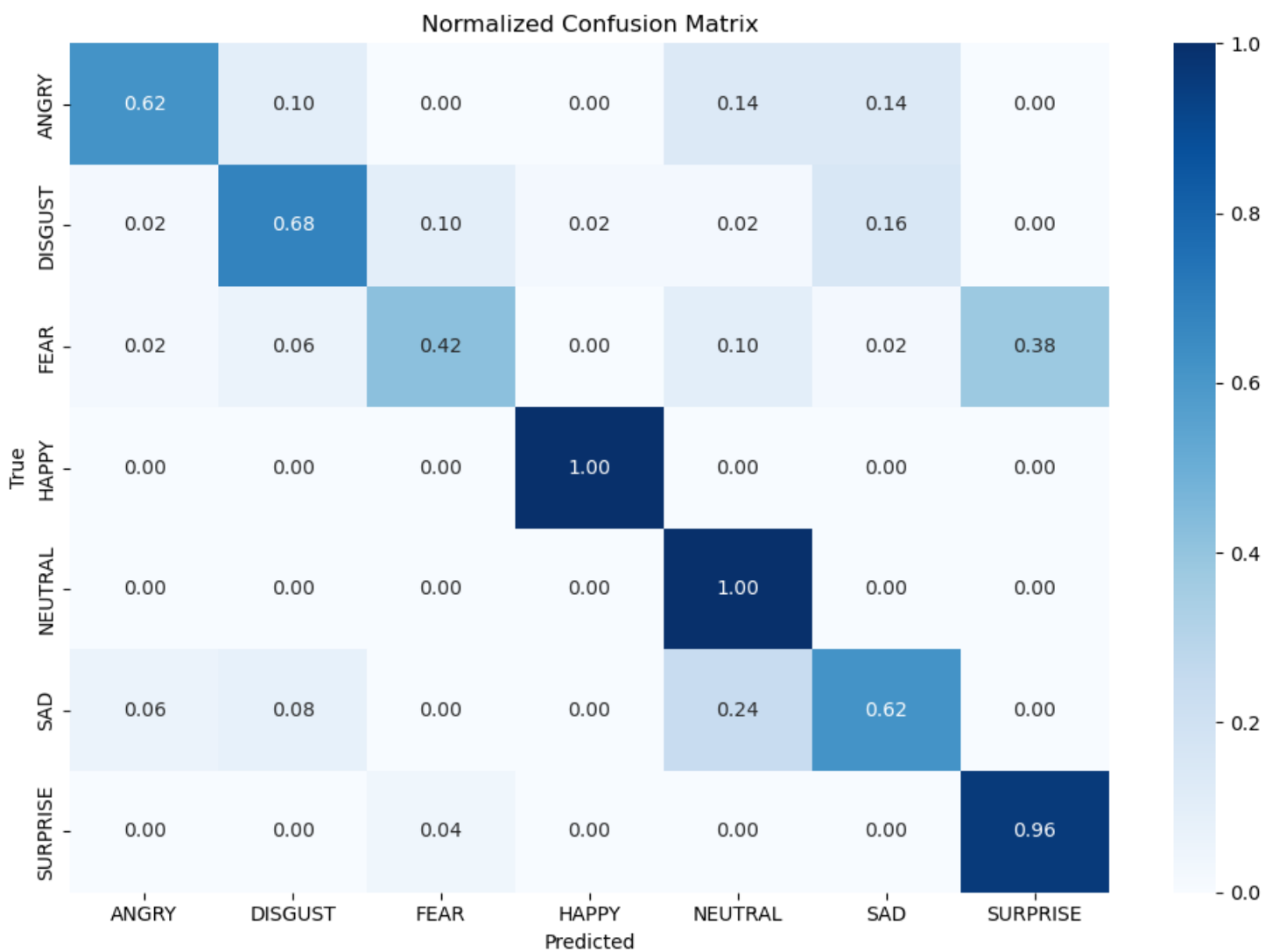


Image: occft_yolo_cm_normalized.png



Images from high_confidence_errors:

Image: high_conf_error_1.png

Pred: DISGUST
Conf: 1.00
True: ANGRY



Pred: DISGUST
Conf: 0.99
True: ANGRY



Pred: DISGUST
Conf: 0.99
True: ANGRY



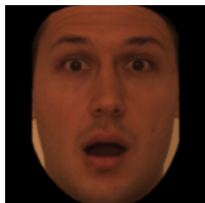
Pred: SURPRISE
Conf: 0.99
True: FEAR



Pred: NEUTRAL
Conf: 0.99
True: SAD



Pred: SURPRISE
Conf: 0.98
True: FEAR



Pred: NEUTRAL
Conf: 0.98
True: SAD



Pred: NEUTRAL
Conf: 0.97
True: SAD



Pred: NEUTRAL
Conf: 0.96
True: FEAR



Pred: NEUTRAL
Conf: 0.95
True: SAD



Pred: SURPRISE
Conf: 0.94
True: FEAR



Pred: SURPRISE
Conf: 0.94
True: FEAR



Pred: NEUTRAL
Conf: 0.93
True: SAD



Pred: FEAR
Conf: 0.92
True: DISGUST



Pred: SAD
Conf: 0.92
True: DISGUST



Pred: SAD
Conf: 0.91
True: ANGRY



Pred: SURPRISE
Conf: 0.89
True: FEAR



Pred: NEUTRAL
Conf: 0.89
True: SAD



Image: high_conf_error_2.png

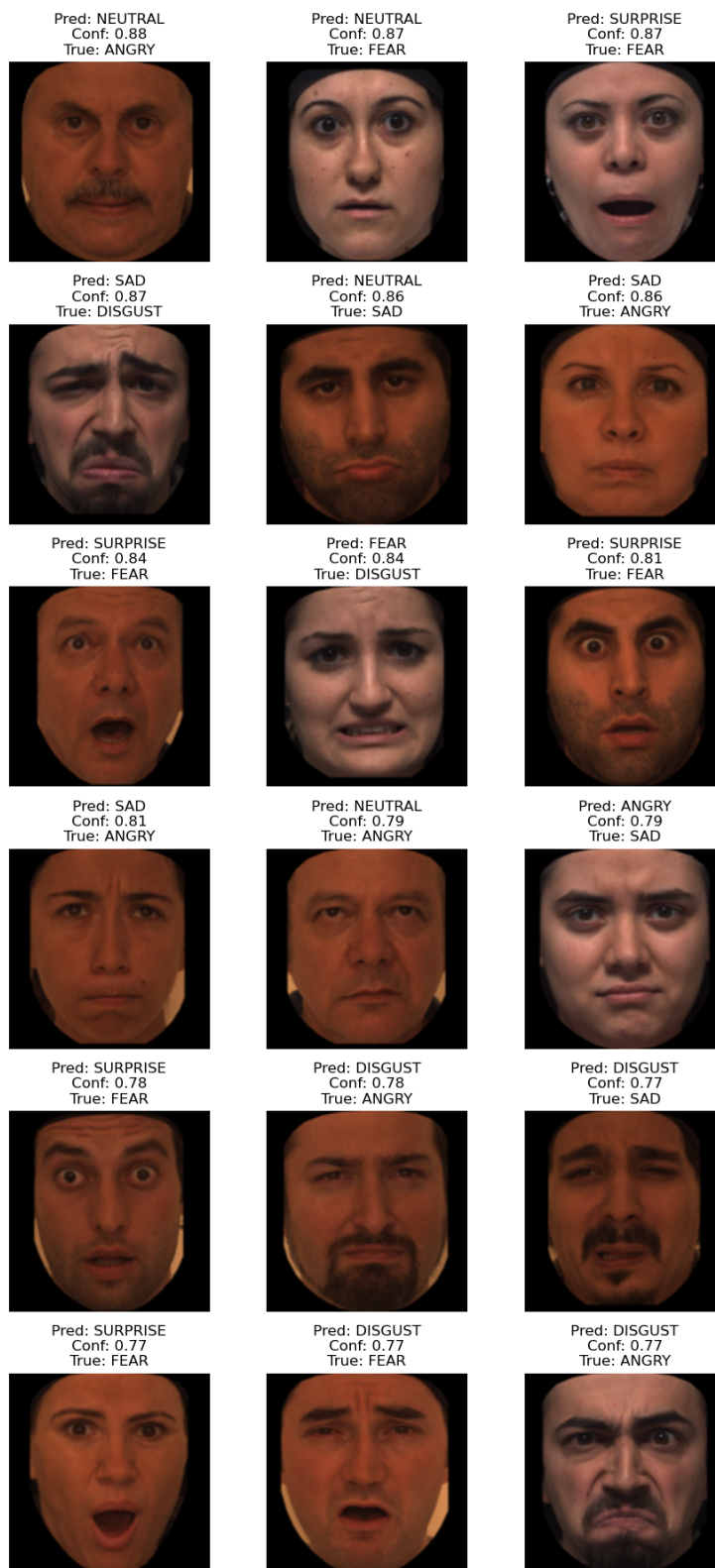
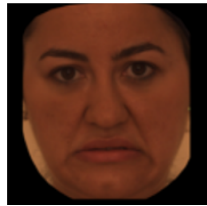


Image: high_conf_error_3.png

Pred: SAD
Conf: 0.76
True: DISGUST



Pred: HAPPY
Conf: 0.74
True: DISGUST



Pred: SAD
Conf: 0.72
True: DISGUST



Pred: NEUTRAL
Conf: 0.66
True: SAD



Pred: SURPRISE
Conf: 0.63
True: FEAR



Pred: DISGUST
Conf: 0.62
True: SAD



Pred: SAD
Conf: 0.76
True: ANGRY



Pred: SAD
Conf: 0.73
True: DISGUST



Pred: FEAR
Conf: 0.71
True: SURPRISE



Pred: SURPRISE
Conf: 0.65
True: FEAR



Pred: NEUTRAL
Conf: 0.63
True: ANGRY



Pred: NEUTRAL
Conf: 0.62
True: DISGUST



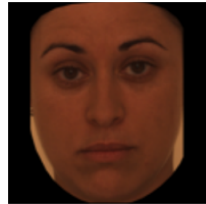
Pred: SURPRISE
Conf: 0.75
True: FEAR



Pred: SAD
Conf: 0.73
True: DISGUST



Pred: NEUTRAL
Conf: 0.68
True: SAD



Pred: SURPRISE
Conf: 0.64
True: FEAR



Pred: NEUTRAL
Conf: 0.63
True: ANGRY



Pred: NEUTRAL
Conf: 0.62
True: ANGRY



Image: high_conf_error_4.png

Pred: SAD
Conf: 0.60
True: DISGUST



Images from tsne:

Image: tsne_occft_yolo_perplexity_20.png

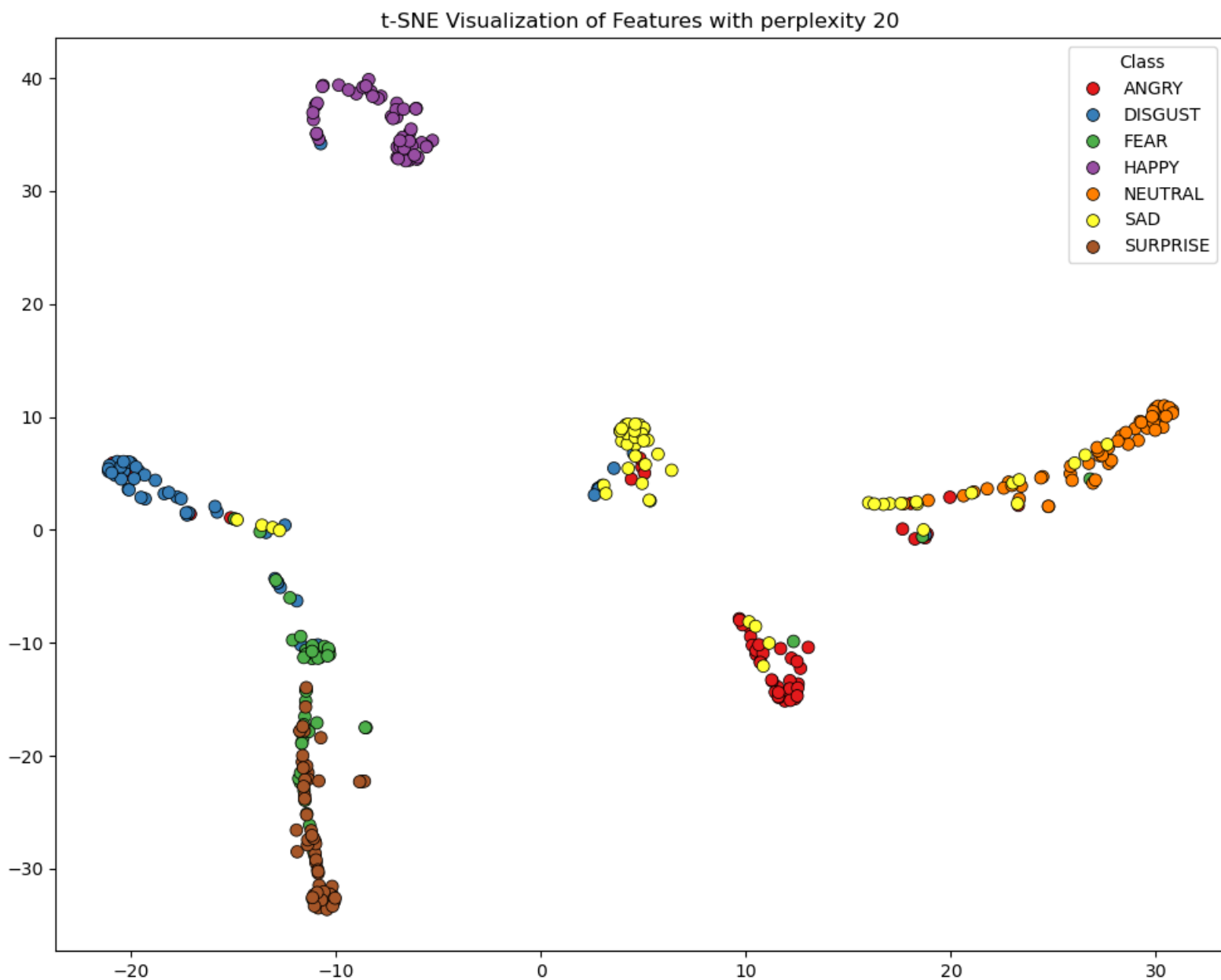


Image: tsne_occft_yolo_perplexity_30.png

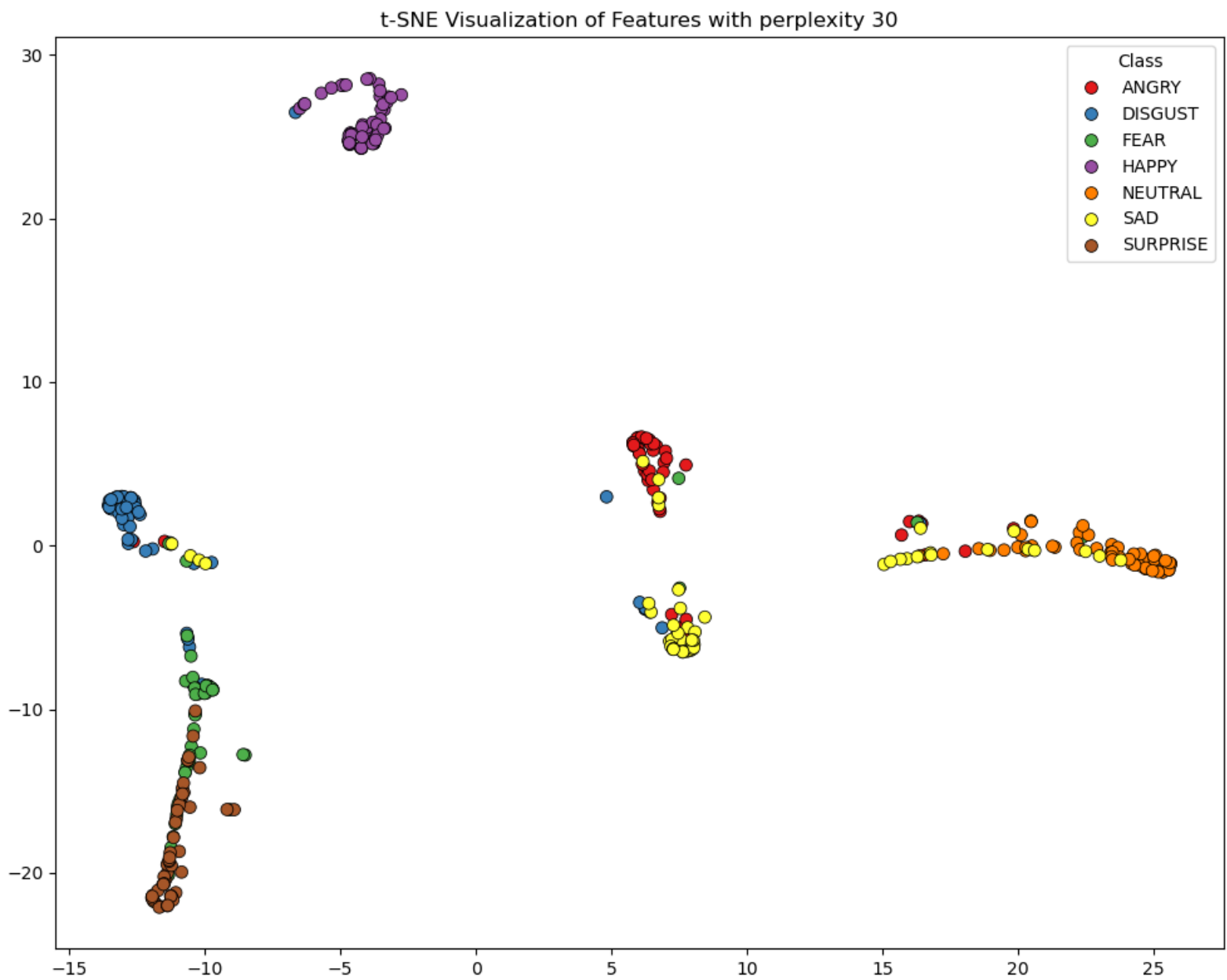


Image: tsne_occft_yolo_perplexity_5.png

t-SNE Visualization of Features with perplexity 5

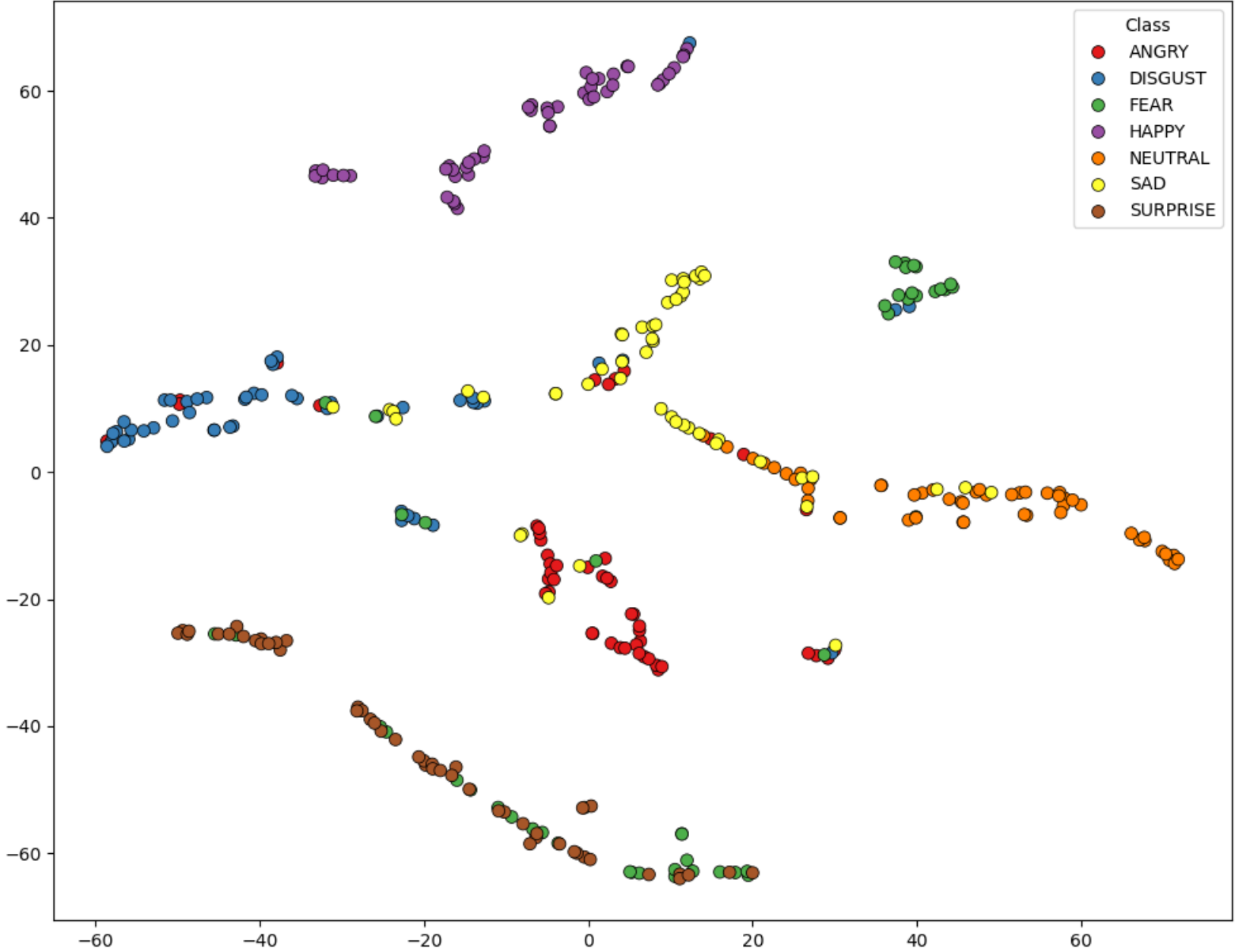
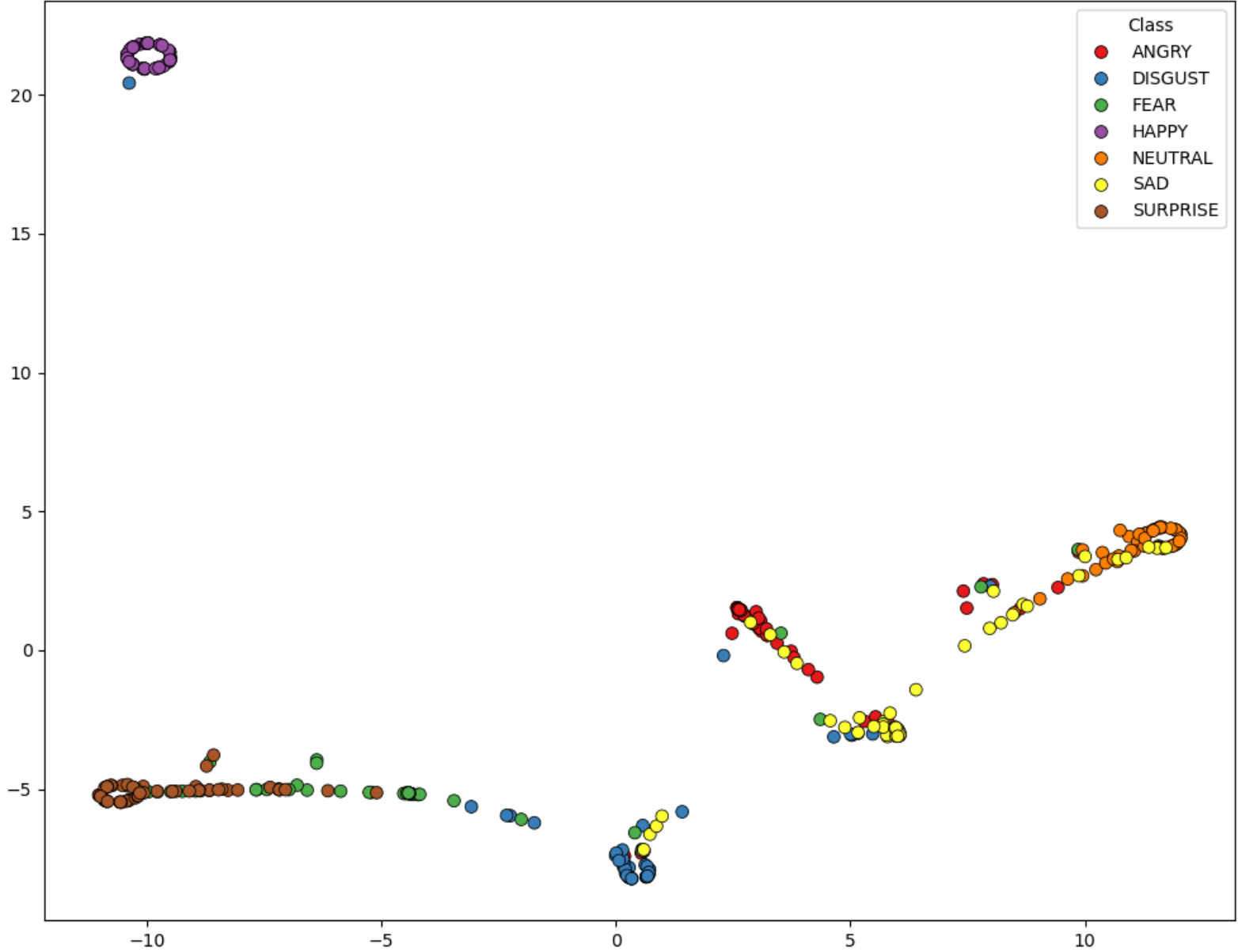


Image: tsne_occft_yolo_perplexity_50.png

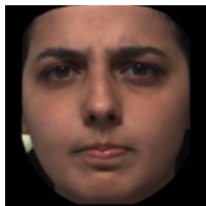
t-SNE Visualization of Features with perplexity 50



Images from uncertain_predictions:

Image: uncertain_predictions_1.png

Pred: ANGRY (0.44)
Sec: SAD (0.41)
True: ANGRY



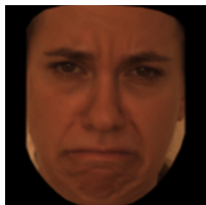
Pred: SAD (0.50)
Sec: ANGRY (0.47)
True: ANGRY



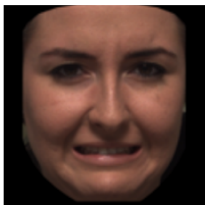
Pred: NEUTRAL (0.46)
Sec: ANGRY (0.41)
True: ANGRY



Pred: SAD (0.51)
Sec: DISGUST (0.49)
True: DISGUST



Pred: FEAR (0.42)
Sec: DISGUST (0.39)
True: DISGUST



Pred: FEAR (0.50)
Sec: DISGUST (0.44)
True: DISGUST



Pred: ANGRY (0.40)
Sec: FEAR (0.34)
True: DISGUST



Pred: FEAR (0.53)
Sec: SURPRISE (0.47)
True: FEAR



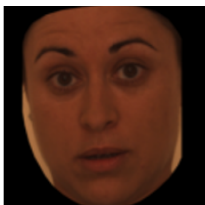
Pred: DISGUST (0.51)
Sec: FEAR (0.45)
True: FEAR



Pred: SURPRISE (0.46)
Sec: FEAR (0.43)
True: FEAR



Pred: NEUTRAL (0.31)
Sec: FEAR (0.31)
True: FEAR



Pred: SURPRISE (0.53)
Sec: FEAR (0.46)
True: FEAR



Pred: NEUTRAL (0.52)
Sec: SAD (0.47)
True: SAD



Pred: SAD (0.48)
Sec: NEUTRAL (0.46)
True: SAD



Pred: SAD (0.45)
Sec: ANGRY (0.37)
True: SAD



Pred: ANGRY (0.44)
Sec: SAD (0.36)
True: SAD



Pred: SURPRISE (0.54)
Sec: FEAR (0.46)
True: SURPRISE

